



The Short-Term Effects of COVID-19 on the Manufacturing Industry in Colombia: Unveiling Insights through a Natural and Exogenous Experiment

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Abstract

This study analyzes the economic impact caused by the beginning of the COVID-19 pandemic, characterized by the confinement in the second and third quarters of 2020, on the Colombian manufacturing industry. It implemented a Difference-in-Difference (Dif-in-Dif) set-up model design; in this case, it measures the impacts of the industrial production index on the Colombian manufacturing industry, employment, manufacturing exports, and imports, considering the COVID-19 crisis as a random natural experiment. For this, the Dif-in-Dif methodology was used for time series data and variables to describe the behavior progress of COVID-19 in Colombia. The results show that the COVID-19 pandemic has had a statistically significant impact on the industrial manufacturing production index, the number of employed individuals in the manufacturing sector, and the manufacturing of imports, which signifies a negative impact concerning the number of individuals who tested positive for COVID-19 and the number of deaths and recoveries.

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Introduction

Late in 2019, in Wuhan, Hubei province, the epidemic outbreak of COVID-19 was discovered, and the number of infected individuals did not exceed 100. However, this number rapidly increased in China and worldwide due to the disease's easy transmissibility and the current global development. Therefore, the World Health Organization (WHO) indicated the risks to public health worldwide. Additionally, by January 30, 2020, in China, the number of infected people had already exceeded 9,000; by that time, there were 20 cases outside China. Then, the WHO declared the COVID-19 crisis as a global health emergency. Governments adopted drastic measures to tackle the spread of the virus because of the high transmissibility of COVID-19. However, these measures seriously slowed down the world economy. Therefore, a series of economic problems appeared, causing an adverse effect. For example, in March 2020, in the United States, at least 22 million people requested government subsidies to offset the social crisis due to the economic shutdown caused by COVID-19.

On March 6, 2020, the Ministry of Health confirmed the first coronavirus case in Colombia. Among the contingency plans adopted by the government and consistent with worldwide measures, the home isolation policy was in effect from March 22 to August 31, 2020. As a result, many sectors of the economy had to cease operations, affecting the labor supply and the behavior of the industries. This situation implied changes in production to prevent the spread of the disease. All this added to the high volatility of the dollar, the depreciation of the exchange rate, and the drop in crude oil prices, a set of problems that the Colombian government had to tackle. Additionally, unemployment experienced a significant change rising from 12.6% (March 2020) to 21.4% (May 2020).

This article aims to assess the impact of the arrival of the COVID-19 pandemic in Colombia on the manufacturing industry aggregate. A Difference-in-Differences analysis approach was used to achieve the proposed objective based on the methodological proposal of natural experiments starting from a random and exogenous fortuitous event coinciding with results of interest in the manufacturing industry.

The COVID-19 pandemic has had a significant impact on the economy and the labor market overall. In a comprehensive study, Cardona-Arenas and Sierra-Suarez (2023) analyze the impact of the pandemic on the persistence of unemployment duration in Colombia. The authors construct an indicator of this persistence and demonstrate that remittances and non-labor income positively affect the long-term unemployment rate (LAPU), especially during the COVID-19 crisis. These findings are explained in the study by noting that remittances and non-labor income can finance longer periods of job search, resulting in an increase in long-term unemployment, even in the post-pandemic period.

The industry plays a fundamental role in the economy and job creation, making it crucial to understand how the crisis has impacted this sector. Studies like those of Ashraf and Goodell (2022) and Brezdeń (2023) emphasize the importance of analyzing the pandemic's effect on the industry, particularly on the value chain, which is crucial for policymakers. For instance, Zhao and Kim (2021) conclude that the COVID-19 crisis has affected textile companies, forcing them to adapt to market changes. In this regard, Queiroz et al. (2020) review valuable resources to mitigate the pandemic's impacts, such as innovation capacity, operational flexibility, accurate information, and sustainability. In summary, analyzing how the pandemic has affected industrial activity provides key insights into a country's economic health. Furthermore, these effects can ripple into other economic sectors, like trade, services, and investment, underscoring the importance of understanding these impacts (see: Ando & Hayakawa, 2022). The reviewed literature supports this notion by highlighting the relevance of studying the pandemic's effects on the industry. By gaining a better understanding of how the crisis has impacted industrial activity, decision-makers can devise more effective strategies to mitigate negative impacts and promote economic recovery. Unfortunately, no studies have been applied to Colombia that fulfill this purpose for the sector in question.

In accordance with the objective of assessing the impact of the COVID-19 pandemic in Colombia on the aggregate manufacturing industry for the period 2020–2024, numerous related studies have been identified. However, few of them offer novel approaches that are of significant interest for the present study. Among these, the following stand out: Brezdeń (2023), who, through the calculation of a synthetic indicator and vector analysis, measures the impact of the COVID-19 pandemic on industrial activity in Poland. On the other hand, Wang (2023) analyzes methodological innovations for measuring impact developed by Archanskaia et al. (2023), applied to a set of industrial companies in the European Union. Both Wang (2023), and Archanskaia et al. (2023) agree that evaluating corporate financial distress based on industrial turnover is novel. However, Wang (2023) emphasizes in their critique that the approach used may be susceptible to the Lucas critique, as the model's ability to predict the behavior of firms in unobserved situations is limited, posing challenges for estimating results at the firm level with the necessary precision.

Overall, to control the COVID-19, the countries have implemented lockdown measures with the sole purpose of restricting population movement. Overall, populations in countries with high institutional quality have been less receptive to lockdown measures due to their economic effects (Ferraresi et al. 2020). For the Asia-Pacific context, it can be concluded that government policies, lockdowns, and travel restrictions implemented during the COVID-19 outbreak have slowed economic growth in the transportation, manufacturing, and agriculture sectors (Bigelow et al., 2021). Ashraf and Goodell (2022) reach similar conclusions in the context of OECD countries, emphasizing that while such policies have a short-term adverse effect, they also have a long-term recovery effect on economic growth.

After conducting an exhaustive literature review, it is noteworthy that none of the identified works to date has addressed an impact evaluation methodology based on the specific characteristics of the crisis, either internationally or in the Colombian context. This implies considering the COVID-19 pandemic as a random, natural, and exogenous experiment, a perspective not previously explored in the field of economic research, at

least without resorting to a set of overly restrictive and sometimes unrealistic assumptions. This novel approach is of great importance, as thus far, the conditions to apply this type of impact analysis have only been met in disciplines such as exact, natural, biological, or health sciences, without relying on excessively restrictive assumptions in the field of economics. Therefore, this work represents a significant advancement and bridges the knowledge gap by addressing this problem from an innovative and rigorous perspective, ensuring the attainment of robust and unbiased results, as will be demonstrated later on.

The analysis of the impact of COVID-19 on the economy highlights a significant economic slowdown in various industries, evidenced by studies exploring the effects of lockdown policies and pandemic-related uncertainty on industrial investment plans (Ashraf & Goodell, 2022; Chen et al., 2021). This decrease in investment plans could be attributed to the uncertainty and restrictions imposed by the health emergency (Pasa et al., 2020). However, it is important to consider that government economic recovery policies, while potentially effective, have also coincided with an increase in the number of deaths, underscoring the importance of implementing strict biosafety protocols (Gong et al., 2020). Furthermore, recent studies have emphasized the need for a more comprehensive evaluation of the economic impacts of the virus, exploring aspects such as economic growth, industrial activity, exchange rate behavior, and other financial indicators (Cardona-Arenas & Serna-Gómez, 2020; Leduc & Liu, 2020).

The current uncertainty due to the lack of empirical evidence from similar situations has caused multiple studies to be conducted within the international context to show the possible economic and social consequences caused by the pandemic. Among these studies, the study by Bonet Morón, et al. (2020) shows the economic impacts on the Colombian economy from the prolonged preventive isolation. This isolation may have caused losses between 0.6% and 6.1% of the national GDP, affecting mainly the economic branch of services such as the hotel, tourism, and manufacturing sectors which are directly impacted by the significant variations in domestic and foreign trade, among other factors. González et al. (2020) reach similar conclusions in their economic impact analysis caused by COVID-19 in Galicia.

This paper aims to study the impact of COVID-19 in the manufacturing industry on aggregate variables of the sector. For this purpose, a literature review was conducted to support the findings conceptually. Then, the methodology of the difference-in-difference set-up associated with a natural-random and exogenous experiment is presented to estimate a model representing the significant effects of the industrial production index changes in response to the COVID-19 crisis. Finally, the results obtained from the model and the study's general conclusions are presented.

Literature Review

The manufacturing industry plays a fundamental role in economic analysis, recognized for its positive effects on economic growth and productivity (Osorio et al., 2022). Kaldor (1966) presents his model of economic growth focused on the importance of industrial-manufacturing growth. From there, Kaldor (1966, 1967, 1968, 1970) establishes his three fundamental laws of economic growth: first, industrial-manufacturing production drives economic growth and increases GDP. Second, the growth of labor productivity in this sector is positively correlated with production.

Third, the growth of total economic productivity is positively associated with manufacturing industrial production, affecting overall productivity.

In the analysis of the determinants of productivity in the manufacturing industrial sector, a decrease in the participation of these industries in advanced economies is observed due to post-industrialization (Rodrik, 2016). Despite the need for government plans to boost this industry (Feraud & Flores, 2019), these are questioned by Little et al. (1970). Additionally, the importance of human capital in the technical efficiency of the sector is emphasized (Valderrama et al., 2015). In the specific case of Colombia, Flórez and Ramírez (2016) found inelasticity in import demand in certain manufacturing sectors. Castells and Schorr (2015) also observed shortcomings in the reindustrialization model in Argentina.

State policies implemented during the COVID-19 crisis, particularly in the manufacturing sector, helped mitigate economic repercussions. Fiscal measures, like tax payment deferrals, reduced costs for companies. Additionally, incentives in the construction sector, such as lower mortgage rates and financial support for housing, boosted construction activity and demand for materials (ECLAC, 2012). Governmental efforts notwithstanding, the primary cause of the notable economic downturn was confinement measures. To narrow the research focus, searches were conducted across databases from 2020 to 2024. Two main areas emerged: reviews and assessments of potential economic impacts, and impact estimations. Prominent publications covered areas like shipping data, global supply chains, and electricity consumption.

To understand the impact of COVID-19 on the manufacturing industry, it is essential to review the international literature, highlighting methodological trends. The pandemic has sparked significant interest in understanding how it affects this key sector, leading to numerous studies on its implications. For example, Ashraf and Goodell (2022) and Brezdeń (2023) emphasize the importance of analyzing the impact of COVID-19 on the value chain of the manufacturing industry, crucial for formulating effective policies. Research such as that of Zhao and Kim (2021) demonstrates how the crisis has compelled textile companies to adapt to drastic market changes.

Furthermore, studies like that of Queiroz et al. (2020) identify valuable resources for mitigating the negative effects of the pandemic on the manufacturing industry, such as innovation capacity and operational flexibility. These findings underscore the importance of understanding how the pandemic has impacted industrial activity to assess a country's economic health.

It is crucial to consider how these effects may extend to other economic sectors, such as trade, services, and investment. Ando and Hayakawa (2022) emphasize that understanding these impacts is essential for developing effective economic recovery strategies.

The outbreak of COVID-19 has had alarming effects on human lives and the economies of the affected countries. Due to the prolonged closure of manufacturing centers, the economic impact transcends borders, including territorial borders, through global supply chains (Yu and Aviso, 2020). Although several studies have been conducted on the impact of COVID-19 on the manufacturing industry internationally, it is important to note the lack of specific research applied to countries like Colombia. However, there are some relevant studies that offer novel approaches to understanding the pandemic's impact on industrial activity. For instance, Brezdeń (2023) uses a synthetic indicator and vector analysis to measure the impact on industrial activity in Poland, while Wang

(2023) analyzes innovative methodologies for evaluating corporate financial distress in the European Union.

However, it has been noted that some approaches may be susceptible to methodological criticisms, such as limited ability to predict business behavior in unobserved situations (Wang, 2023). In this regard, Wang (2023) examines methodological innovations developed by Archanskaia et al. (2023), applied to industrial companies in the European Union. Both Wang (2023), and Archanskaia et al. (2023) agree that assessing corporate financial distress based on industrial turnover is novel. However, Wang (2023) highlights that this approach could be susceptible to Lucas' critique due to its limited ability to predict business behavior in unobserved situations, posing challenges for estimating results at the enterprise level with the necessary accuracy.

It is widely known that policymakers have implemented a wide range of policies and jurisdictions to combat the spread of COVID-19, which over time have required a methodological research design of differences in differences (DD) to estimate the causal effects of COVID-19 countermeasures. Goodman-Bacon and Marcur (2020). However, no studies have examined the COVID-19 crisis as an exogenous natural event, allowing for analysis as a random experiment, as undertaken in this study. This perspective aligns with Chernozhukov et al.'s characterization of the COVID-19 crisis as an exogenous natural event (2024). Variables stemming from this event can be considered useful natural experiments for causal identification, even in the presence of unobserved confounding. This assumption posits that the treatment assignment mechanism is purely random. Empirical researchers often use the classical additive linear regression model to infer the Average Treatment Effect (ATE), where covariates enter additively, offering practical advantages and often leading to improvements in precision over the simple two-means approach.

Methodology and Method of Identification

Estimations for identifying the effects of random COVID-19 pandemic events on the Colombian manufacturing industry consisted of calculating the *set-up* difference-in-difference estimator through Ordinary Least Squares (OLS).

For this purpose, we use time series for which the treatment effect considered in this analysis is the number of COVID-19 infections for 2019, when there were no reported cases of the virus in Colombia but in China.¹ Therefore, in Colombia, the first infection was recorded on March 6, 2020; the previous month is considered the pre-treatment period, and from March onward, the post-treatment period. Table 1 summarizes the information needs. Below is an algebraic deduction of the general equation to be estimated, taking into account Bernal and Peña (2011) and Ciani and Fisher (2020).

¹ On January 12, 2020, China made the genetic sequence of the virus that caused COVID-19 public, and by January 13, 2020, a case of COVID-19 was officially confirmed in Thailand, the first case registered outside China.

Table 1 Treatment and Control Periods

	Treatment	Control
t = 1 (baseline)	$(\bar{\gamma}_1 D = 1)$	$(\bar{\gamma}_1 D = 0)$
t = 2 (Follow-up)	$(\bar{\gamma}_2 D = 1)$	$(\bar{\gamma}_2 D = 0)$

Source: Authors' elaboration

A binary variable is thus defined T_i equal to zero in the period before the event-treatment and equal to 1 in the period after the post-event-treatment, so the difference-in-differences estimator over time is given by:

$$\tau = (\bar{\gamma}_1|D = 1) - (\bar{\gamma}_1|D = 0) \quad (1)$$

$\bar{\gamma}_1$ is the average outcome variable in the industrial production index of the manufacturing industry. The estimator can be obtained by the OLS method by estimating:

$$\gamma_t = \mu + \beta_1 T + u_i \quad (2)$$

In this case $\hat{\beta}_1$ corresponds to the difference-in-differences estimator over time; since the COVID-19 event is a random exogenous event, the estimator $\hat{\beta}_1$ is unbiased, the method is appropriate for evaluating a treatment (ongoing event) in the absence of a control group, that for the methodology is not necessary. To calculate differential effects: $(\bar{\gamma}_1|D = 1)$, $(\bar{\gamma}_1|D = 0)$ and interaction: $(\bar{\gamma}_1|D = 1) \forall T_i = 1$

$$\gamma t = \mu + \beta_1 D_{it} + \beta_2 T + \beta_3 (DT) + \epsilon_{it} \quad (3)$$

The identification problem starts from the description of the differential effects using the following equation structure:

$$E(\bar{\gamma}_1|D = 1, T = 1) = \mu + \beta_1 + \beta_2 + \beta_3 \quad (4)$$

$$E(\bar{\gamma}_1|D = 0, T = 1) = \mu + \beta_2 \quad (5)$$

$$E(\bar{\gamma}_1|D = 1, T = 0) = \mu + \beta_1 \quad (6)$$

$$E(\bar{\gamma}_1|D = 0, T = 0) = \mu \quad (7)$$

Furthermore, solving the equation:

$$E(\bar{\gamma}_1|D = 1, T = 1) - E(\bar{\gamma}_1|D = 0, T = 1) = \beta_1 + \beta_3 \quad (8)$$

$$E(\bar{\gamma}_1|D = 1, T = 0) - E(\bar{\gamma}_1|D = 0, T = 0) = \beta_1 \quad (9)$$

$$\gamma t = \mu + \beta_1 D_{it} + \beta_2 T + \beta_3 (D_{it} \times T_{it}) + \delta_i X_{it} + \varepsilon_{it} \quad (10)$$

Thus, the reduced form of the model is:

$$\gamma t = \mu + \beta (D_{it} \times T_{it}) + \delta_i X_{it} + \varepsilon_{it} \quad (11)$$

Subtracting Eq. (8) from Eq. (9) identifies the coefficient of interest, $\hat{\beta}_3$, that is the difference-in-difference estimator. The unbiased OLS estimation guarantees that $E(\varepsilon_{it} | D_{it}, T_{it}) = 0$, the assumption necessary to ensure treatment-event randomization (COVID-19). The expected value of the stochastic error term must be tested. ε_{it} is orthogonal to D_{it} and T_{it} . It must also be proven that D_{it} is orthogonal to the set of observable variables included in the control vector $X : [x_{1t}, x_{2t}, \dots, x_{nt}]$ so that in the model $Dit = \delta_0 + \delta_1 x_{1t} + \delta_2 x_{2t} + \dots + \delta_n x_{nt} + \varepsilon_{it}$, the event-treatment Dit will not be correlated with the observable variables so that coefficients $\delta_1 \dots \delta_n$ will be approximately equal to zero. The F-test will be analyzed for joint significance and to calculate robust standard errors. If the results suggest the absence of randomness, the estimate can be adjusted by introducing additional controls to the vector. Thus, the following equation includes a vector of control variables X_{it} .

$$\gamma t = \mu + \beta (D_{it}) + \delta_i X_{1t} + \delta_i X_{2t} + \varepsilon_{it} \quad (12)$$

It is important to note that the efficiency of the difference-in-difference estimator (Dif-in-Dif) depends on the fraction IPI variance that is explained by unobserved factors.

Variables and Data

According to the purpose of this paper, the behavior of the chosen response variable is analyzed. It is Colombia's industrial manufacturing production index, base year 2018 = 100, and monthly data recovered from the National Administrative Department of Statistics (DANE). There is evidence of a decrease in time consistent with the presence of the pandemic in the Colombian territory, which different factors may cause.

First, the dependence of Colombian production on variations in the international price of oil presents a significant decrease with the arrival of the COVID-19 pandemic in Colombia. Second, during the first months of the pandemic, to mitigate the spread of the virus, national production contracted by 4.1% for the third quarter of 2020 (Departamento Administrativo Nacional de Estadística, 2021). Third, the vector of control variables introduced in the model is composed of variables such as remittances with daily data retrieved from the Bank of the Republic of Colombia, data provided by DANE that collects information such as the economy tracking index, number of people of long-term unemployed sector, employed population of manufacturing industry sector and exports and imports of the same sector.

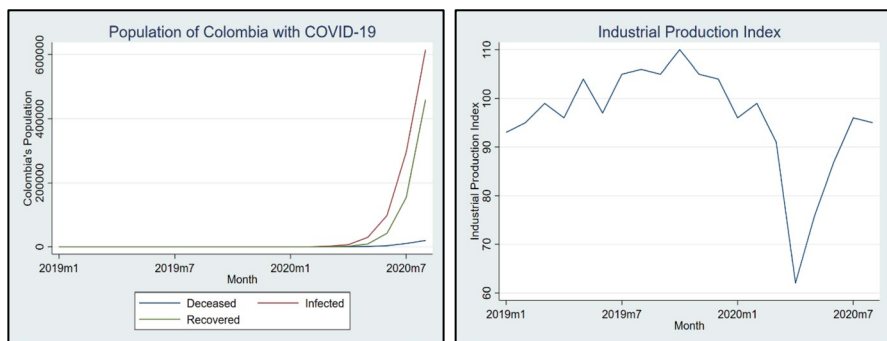
Table 2 Variable Interaction of Follow-up in Colombia

Variable Name	Sample	Calculated Interaction Composed for the Product
Dt_Infections:	2019M1-2020M8	$Dt * (\text{Number of COVID_19 infections})$
Dt_Dead	2019M1-2020M8	$Dt * (\text{Number of COVID_19 infections})$
Dt_Recovered	2019M1-2020M8	$Dt * (\text{Number of COVID_19 recovered})$
Infections_employed	2019M1-2020M8	$(\text{Infections}) * (\text{employed_manufacture})$
Dead_employed	2019M1-2020M8	$(\text{Deaths}) * (\text{employed_manufacture})$

Source: Authors' elaboration

The long-term unemployed (LTU) variable corresponds to question P7320: “How many weeks ago did you last stop working?” for the module on unemployed for the sample of the Large Integrated Household Survey of DANE.² According to DANE, long-term unemployment is described in Colombia as a situation in which a person has been unemployed for 52 weeks without getting a job, meaning a year or more time without working. (DANE, 2020). Alternatively, the Brent daily data benchmark oil price for Colombia is retrieved from *Investing.com*. Finally, interaction variables have been constructed to capture the result of the combined effect on variables of the progress of the pandemic in Colombia. Table 2 shows these variables.

The following is the monthly aggregate of daily data on the evolution of COVID-19 infections, deaths, and recoveries in Colombia from the official database of the National Institute of Health. The data analysis process began by identifying the behavior of the outcome variable before and after the first case registered in Colombia for the COVID-19 pandemic. The first estimated model was the manufacturing industrial production index (IPI) contrasted with the variables that describe the behavior of COVID-19, and subsequently, the model with differential effects was estimated. Figure 1 summarizes the time series of the mentioned variables.



Source: Authors' elaboration.

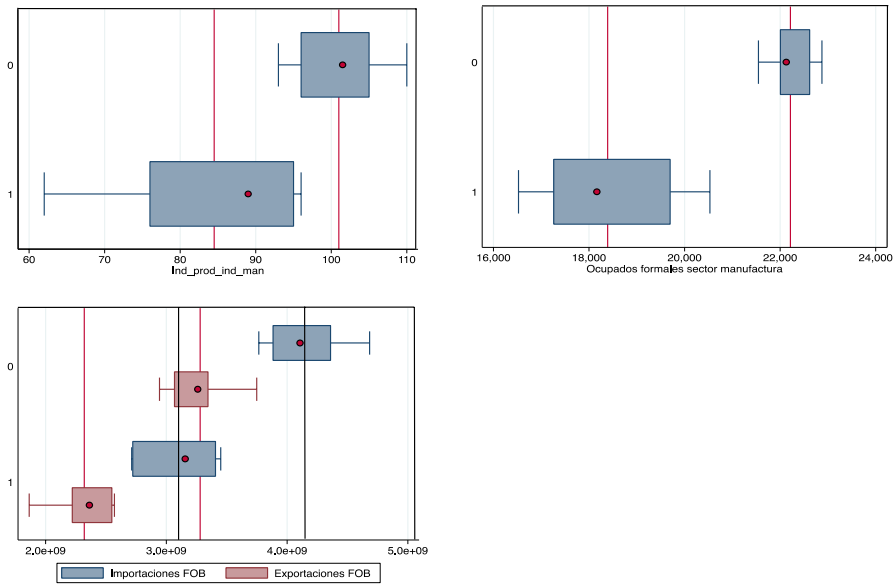
Fig. 1 Time series for Industrial Production Index and Colombia's Population with COVID-19

² The long-term unemployment variable corresponding to question P7320 of the GEIH microdata unemployed module for the headline missed two values for the series in April and March 2020. According to DANE's response, the reports and databases for these months were limited because of the COVID-19 contingency; therefore, DANE does not have this information for the unemployed and inactivity module. For this reason, an imputation was applied to the mean using the NNI (Nearest Neighbor Imputation) method.

Table 3 Statistically Significant Differences in Means of the Variables for t = 1 and t = 2

Variable	Group = 0 t = 1 (baseline)		Group = 1 t = 2 (Follow-up)		Difference	Pr
	Media	Std Dev	Media	Std Dev		
IPI	101	5.15	84.5	13.18	- 16.5***	0.0003
Employed in manufacturing	22,214.14	412.79	18,390.67	1,497.30	- 3,823.476***	0.0000
Manufacturing exports	3,280,000,000	270,000,000	2,320,000,000	264,000,000	- 960,000,000***	0.0000
Manufacturing imports	4,150,000,000	306,000,000	3,100,000,000	356,000,000	- 1,050,000,000***	0.0000

Source: Authors' elaboration based on DANE data



Source: Authors' elaboration based on DANE data.

Fig. 2 Comparison of Means and Medians for Variables for $t=1$ and $t=2$. Note: the red vertical line corresponds to the mean of the variable corresponds to the period $t=1$: Baseline corresponding to pre-treatment period (group=0) and $t=2$: Follow-up corresponding to the post-treatment period (group=1) see Table 1. The red dot corresponds to the median indicator in the same way for $t=1$ and $t=2$. In the case of imports, the vertical black lines represent the mean for $t=1$ and $t=2$

Table 3 shows the statistical results to capture significant differences in means of key manufacturing variables, and Fig. 2 summarizes the comparison of means and medians.

Results and Discussion

After estimating the models and considering all the methodological specifications mentioned, it can be argued that the random shock generated by the COVID-19 pandemic has unleashed several impacts on the manufacturing sector. Indeed, four estimations were carried out that were similar in methodological aspects. However, all the models involve a different analysis for the manufacturing sector. First, a model that provides evidence of the impact generated by COVID-19 on the IPI was estimated. The second model shows the impact of COVID-19 on the number of formal employees in the manufacturing sector; finally, two models explain the impact of COVID-19 on exports and imports in the manufacturing industry.

Concerning these last two models, evaluating the differentiated effects on the sector's current account variables was considered since it is possible that the aggregate result by trade balance does not provide enough information regarding

Table 4 Regressions Calculated through OLS for *Dif-in-Dif*

Robust Linear Regressions by OLS for Difference-in-differences						
Dependent Variable	Model 1: COVID Effect on IPI					
	N. Observations = 20			R-square = 0,9500		
	F(4, 14) = 0					
	Prob > F = 0			Root MSE = 2.9017		
Independent Variable	Coefficient	Robust Std Error	t-stat	P-value	Inf 95%	Sup 95%
Dt	-11.93***	2.906537	-4.11	0.001	-18.16962	-5.701812
Log_Remittances	24.4***	4.225317	5.76	0.000	15.2891	33.41391
Log_ise	51.32**	18.95716	2.71	0.017	10.66078	91.97891
Infections_Employed	-0.0000000217***	2.21E-09	-9.82	0.000	-2.65E-08	-1.70E-08
Deceased_Employed	0.000000699***	7.17E-08	9.74	0.000	5.45E-07	8.52E-07
Constant	-270.4226***	75.93191	-3.56	0.003	-433.2804	-107.5649
Dependent Variable	Model 2: Effect of COVID on the Number of Employed in the Manufacturing Industry					
	Number of obs = 20			R-squared = 0.9571		
	F(5, 14) = 85.96					
	Prob > F = 0.0000			Root MSE = .02421		
Independent Variable	Coefficient	Robust Std Error	t-stat	P-value	Inf 95%	Sup 95%
Dt	-0.1260505***	0.022148	-5.69	0.000	-0.1735531	-0.0785479
Log_Remittances	0.2225253***	0.0409744	5.43	0.000	0.134644	0.3104067
Log_ltu	0.1403538**	0.0580026	2.42	0.030	0.0159507	0.264757
Dt_Infections:	-0.000000528***	1.73E-07	-3.05	0.009	-8.99E-07	-1.57E-07
Dt_Recovered	0.000000724***	2.21E-07	3.27	0.006	2.49E-07	1.20E-06
Constant	7.885261***	0.3684914	21.4	0.000	7.094926	8.675597
Dependent variable	Model 3: Effect of COVID on Manufacturing Industry Exports					
	Number of obs = 20			R-squared = 0.6817		
	F(5, 14) = 2376.96					
	Prob > F = 0.0000			Root MSE = 65879		
Independent variable	Coefficient	Robust Std Error	t-stat	P-value	Inf 95%	Sup 95%
Dt	-60669.86	60101.24	-1.01	0.330	-189574.2	68234.47
Log_ise	571780.4**	266519.4	2.15	0.050	153.0935	1143408
Dt_Infections:	-25.0565***	7.196032	-3.48	0.004	-40.49046	-9.622549
Dt_Deceased	643.4426***	176.7996	3.64	0.003	264.2452	1022.64
Dt_Recovered	6.040712***	2.000144	3.02	0.009	1.750829	10.33059

Table 4 (continued)

Dependent variable	Model 3: Effect of COVID on Manufacturing Industry Exports					
	Number of obs = 20			R-squared = 0.6817		
	F(5, 14) = 2376.96					
	Prob > F = 0.0000			Root MSE = 65879		
Independent variable	Coefficient	Robust Std Error	t-stat	P-value	Inf 95%	Sup 95%
Constant	-2007429	1248054	-1.61	0.130	-4684239	669380.9
Dependent variable	Model 4: Effect of COVID on Manufacturing Industry Imports					
	Number of obs = 20			R-squared = 0.8669		
	F(5, 14) = 75.25					
	Prob > F = 0.0000			Root MSE = 2.0e + 05		
Independent variable	Coefficient ^a	Robust Std Error	t-stat	P-value	Inf 95%	Sup 95%
Dt	-667705***	155002	-4.31	0.001	-1000151	-335258.9
Log_is	1765283**	784643.2	2.25	0.041	82390.74	3448175
Dt_Infections:	-91.60713***	19.08973	-4.8	0.000	-132.5505	-50.66373
Dt_Deceased	2406.911***	473.2	5.09	0.000	1391.998	3421.824
Dt_Recovered	20.3174***	5.197823	3.91	0.002	9.169183	31.46563
Constant	-4952508	3658550	-1.35	0.197	-1.28E + 07	2894301

Source: Authors' elaboration

*is statistically significant at 10%. We estimated the regression equation with heteroscedasticity-robust errors; **is statistically significant at 5%; ***is statistically significant at 1%

^aIt's important to highlight that even though the interaction effects of infections, deaths, or recoveries are relatively low in model 1 and 2, to some extent, these are statistically significant and contribute to controlling the estimated impact, thereby enhancing the statistical robustness and unbiasedness of the model. In models 3 and 4, the magnitude of the effect is significantly greater

the impact on trade flows in the industrial manufacturing sector (see Table 6). Table 4 contains information that corresponds to the estimation of each of the estimated models. Subsequently, the results were interpreted, considering the difference-in-difference coefficients and the model specification.

Model 1: Impact of COVID-19 on IPI

In this first specification, the dependent variable corresponds to the manufacturing production index; the independent variables are the treatment variable (DT), the logarithm of remittances, and the logarithm of the economic tracking index; there is an interaction variable between the number of COVID-19 infections and the number of formal employees in the manufacturing sector; and finally, there

is another interaction variable between the number of COVID-19 deaths and the number of formal employees in the manufacturing sector. The model for this is represented as follows based on the methodology presented:

$$IPI_t = \mu + \beta_1(DT) + \beta_2(Log_{remittances_{2t}}) + \beta_3(Log_{ise_{3t}}) + \beta_4(Infecteds_Employed_{4t}) + \beta_5(Deceased_Employed_{5t}) + \varepsilon_{it} \quad (13)$$

Regarding the DT treatment variables, the results show that after the arrival of COVID-19 in Colombia, the manufacturing industrial production index declined -11.93571 index points due to this exogenous random disturbance. Similarly, the interaction variable Infecteds Employed has a negative expected sign; the coefficient obtained for this variable shows that if the interaction between the number of formal employees in the manufacturing sector and the number of COVID-19 infections increases by one unit, the industrial production index falls by -2.17E-08: -0.00000217 index points; the effect is small but consistent, explained by the reduction in the productive capacity of the employed labor force.

Alternatively, the loss of the business network caused by the decrease in internal and external consumption and the lack of raw materials are among the events that generated the negative impact on the IPI. Another result of interest obtained in the model corresponds to the effect of the interaction variable Deceased_Employed, whose expected sign is positive, specifically, an increase of one unit in the interaction between the number of formally employed and the number of deaths per COVID-19 generates an increase of 6.99E-07: 0.000699 index points – IPI. This result is related to the rise in the number of deaths from COVID-19 and the level of formal employment as there is a gradual return to work activities as the pandemic progresses. The previous has led to the relaxation of confinement measures, which has generated a more significant number of deaths from COVID-19 and a return of employed people to their jobs; the latter is due to the nature of manufacturing labor functions where there is no room for teleworking or remote work in its aggregate. In addition, the economies of the world and Colombia's main trading partners also relaxed their confinement measures, which allowed for a relative improvement in trade flows.

Looking at the coefficients, the Log_remittances variable indicates that 1% increases in remittances generate growth of $(0.01 * 0.243515)$: 0.0024 points³ in the manufacturing industrial production index, presumably due to two relevant factors: the stimulus generated by remittances on aggregate demand, thus increasing the consumption of different manufactured goods; and remittances can constitute a financing mechanism for investment in family manufacturing companies, as proven in the context of European developing countries (see Drinkwater et al., 2003). Alternatively, a 1% increase in the economic tracking index (ISE) positively impacts the IPI corresponding to $(0.01 * 0.5131)$: 0.0051 IPI points. As is known, the ISE assesses the current state of the real economy since it is based on a short-term indicator;

³ The interpretation of the coefficient is understood as a semi-elasticity in points of the response variables consistent with a robust log-level OLS estimator: $\frac{B_i}{100} = \frac{\frac{\Delta y}{y}}{100(\frac{\Delta x}{x})} = \frac{\Delta y}{\% \Delta x}$.

therefore, the fact that it follows positive and increasing paths indicates the correct performance of the country's productive activities. The constant improvement of this index will imply the long-term cumulative causation, including virtuous circles of accumulation capacities, coordination economies (due to the interaction of economies of scale), and market size (demand); specifically, the cumulative causation proposed by Kaldor (1961) is considered (see recent evidence of the aforementioned in Almosabbah and Almoree (2018)).

Model 2: Impact of COVID-19 on the Number of Formal Employees in the Manufacturing Industry

For this model, the dependent variable is the logarithm of the number of employees in the formal sector of the manufacturing industry, as follows:

$$OC_t = \mu + \beta_1(DT) + \beta_2(Log_{remittances_{2t}}) + \beta_2(Log_{ltu_{2t}}) + \beta_3(Dt_Infections_{3t}) + \beta_4(Dt_recovered_{4t}) + \varepsilon_{it} \quad (14)$$

Explanatory variables such as the logarithm of remittances, the Dt_Contagios: and the Dt_Recovered variables, and the logarithm of the number of long-term unemployed were considered as independent variables. This last variable is of interest in this model since the processes of economic expansion or contraction have a significant impact on the long-term unemployment rate (Girardi et al., 2018); therefore, through this variable capturing the effect of the crisis in the cycle on a labor market variable that is directly or indirectly related to the level of employment is possible. Blanchard and Portugal (2001) support this fact, and Blanchard (2018) states that unemployment and its duration tend to absorb the effects derived from temporary shocks that impact the current state of the labor market, for example, a loss of the employability of individuals due to the COVID-19 pandemic.

Thus, the results provide clear evidence that after the notification of the first positive case for COVID-19 in Colombia, there was a negative average impact of $(-0.1260505) \times 100^4$ on the number of formal employees in the manufacturing sector; in other words, for every 100 formal workers in the manufacturing industry, an average of 12.6 workers lost their jobs as a result of the pandemic. This aspect is attributed to the loss of productive capacity due to a deterioration in the labor force caused by confinement measures and economic contraction.

Alternatively, a 1% increase in the logarithm of the value of remittances received is related to an increase of 0.2225%⁵ in the number of formally employed in the manufacturing sector; this could be explained from different perspectives. Sharma and Cárdenas

⁴ In this case, an interpretation is considered as a semi-elasticity of the robust OLS estimator Dif-in-Dif of a log-level model, in which: $100 * \beta_i = \frac{100 * (\frac{\Delta y}{y})}{\Delta x} = \frac{\% \Delta y}{\Delta x}$.

⁵ In this case, an interpretation is considered as an elasticity of the robust OLS estimator of a log-level model, where: $\beta_i = \frac{100 * (\frac{\Delta y}{y})}{100 * (\frac{\Delta x}{x})} = \frac{\% \Delta y}{\% \Delta x}$.

(2018) found that for 31 Mexican states, during the period 2005–2013, remittances provided households with the necessary funds to start family businesses. In addition, they indicated that establishing domestic companies was also likely to reduce the duration of unemployment and increase the employment of more people who are productively dedicated to family businesses. However, this effect must be tested for the small manufacturing companies sector. Additionally, a 1% increase in the long-term unemployed is related to the rise of 0.14 percentage points in the logarithm employed in the industrial manufacturing sector. This relationship is interesting as it could be explained by the fact that during March and June 2020, the LTU presented lower values than previous periods. This fact could be caused by a transition of many long-term unemployed to inactivity due to the confinement measures that prevented them from looking for work or the contraction of the aggregate demand that caused a significant drop in available jobs, especially in the economy's service sector.

Higher levels of labor supply and lower prices possibly translate into higher real wages, which motivates the most skilled labor force to remain in the labor market or to enter formal manufacturing activities, as predicted in the past (Flatau et al., 1991) for the Australian context during the period 1967–1989. A more recent precedent can be found in (Rodriguez-Gil, 2018), who argues that labor benefit shocks tend to raise long-term unemployment permanently but, at the same time, create incentives for people in informal conditions, if they have specific qualifications, to seek jobs in more formal labor sectors and achieve labor protection, especially in times of economic crisis.

The variable $Dt_Contagios$ had the expected impact on the logarithm of the number of formal employees in the manufacturing sector; this is because the $Dt_Contagios$ variable shows the relationship between the treatment variable and the number of COVID-19 infections, which would lead one to expect a negative coefficient since the increase in the number of infections led to the intensification of confinement and isolation measures, explanatory aspects of the economic contraction. The coefficient of -0.0000528 indicates that the increase in the number of COVID-19 infections implies a reduction in the number of people employed by $100*(-0.0000528)$: -0.0052 ; although the effect is small, it is statistically significant. Regarding the impact of the number of people recovered from COVID-19 on the logarithm of the number of formally employed, the relationship was positive, which is explained by the fact that as the pandemic progressed. The containment measures were progressively relaxed for certain productive sectors; the IPI experienced, on average, a positive impact of $100*(0.0000724)$: 0.00724 in the number of employed people in the industry.

Model 3: Effect of COVID-19 on Manufacturing Industry Exports

For this model, the logarithm of manufacturing industry exports is considered a dependent variable, and as independent variables: i. the treatment variable Dt , ii. the logarithm of the economy's tracking index, iii. the variable $Dt_Contagios$, $Dt_Deceased$ and $Dt_Recovered$, as follows:

$$Ex_t = \mu + \beta_1(DT_{1t}) + \beta_2(Log_{ise2t}) + \beta_3(Dt_Infections_{3t}) + \beta_4(Dt_Dead_{4t}) + \beta_5(Dt_recovered_{5t}) + \varepsilon_{it} \quad (15)$$

The result obtained from the estimation of this model indicates that the treatment variable *Dt* does not have a statistically significant impact on the logarithm of the value of manufacturing exports. However, the estimated coefficient has the expected sign, indicating that since the arrival of COVID-19, the logarithm of manufacturing sector exports has contracted considerably. However, the economic tracking index shows a positive relationship with the logarithm of exports, i.e., a 1% increase in the economic tracking index is significantly related to an increase in the logarithm of exports.

Observing the impact of the interaction variable *Dt_Infections* on the logarithm of imports reveals that the measures adopted by the Colombian government to avoid the spread generated this decrease in the exports of the manufacturing sector. This result makes sense insofar as the confinement had a direct impact on the economic dynamics of the country; the flow of exports would be lower from the supply side – GDP. Therefore, the increase in one unit in the variable *Dt_Infections* due to the virus is related to a significant decrease in $100 \times (-25.0565)$: 2505.6 dollars FOB of the country's manufactured industrial exports. Alternatively, the interaction variable with the number of deaths from COVID-19 has a similar impact to the impact it had in the first estimated model since, as mentioned above, the increase in the number of deaths from COVID-19 is associated with greater contagion and greater flexibility of the containment measures. In turn, the economy's gradual reactivation process explains that there has been a positive impact on exports in the manufacturing sector by $100 \times (643.4426)$: 643,344.26 FOB dollars. The interaction variable *Dt_Recovered* considers the number of recoveries provided and evidence of the expected impact since, as the number of recoveries per COVID-19 increased, Colombian industries aligned with the national guidelines and resumed their productive activities more intensively, which increased the flow of foreign trade from the supply side. According to the results obtained, the increase in the number of recovered products by one person increases the logarithm of exports by $100 \times (6.040)$: 604.07 FOB dollars.

Model 4: Effect of COVID-19 on Manufacturing Industry Imports

For the last model, the logarithm of the manufacturing industry's imports is considered the dependent variable and independent variables: i. the treatment variable *Dt*, ii. the logarithm of the economy's tracking index, iii. the variable *Dt_Contagios*, *Dt_Deceased* and *Dt_Recovered*, as follows:

$$Im_t = \mu + \beta_1(DT_{1t}) + \beta_2(Log_{ise2t}) + \beta_3(Dt_Infections_{3t}) + \beta_4(Dt_Dead_{4t}) + \beta_5(Dt_recovered_{5t}) + \varepsilon_{it} \quad (16)$$

The treatment variable *Dt* also negatively impacted the logarithm of imports by $100 \times (-667705)$: -66,770 dollars FOB. In the present estimated model, as in the previous model, the logarithm of the index of economic monitoring *ISE* shows a positive

Table 5 Synthesis of Impact Coefficients Dif-in-Dif

Variable Results	Treatment Dt	p-value	Constant	p-value
Impact in IPI	-11.9357***	0.001	-270.4226***	0.003
Impact in Occupied	-0.1260***	0.000	7.88526***	0.000
Impact in Exports	-60669	0.330	-200742	0.130
Impact in Imports	-66770***	0.001	-495250	0.197

Source: Authors' elaboration

*statistically significant at 10%. The difference-in-differences estimator was the result of estimating regression equations with heteroscedasticity-robust errors; **statistically significant at 5%; ***statistically significant at 1%

and significant relationship with the logarithm of imports, given that improving conditions in the real economy led to an increase in trade flows, as predicted by Tinbergen (1963) as a precursor of the gravity model of trade and more recently for the Colombian context by Bolivar et al. (2019). The interaction variable *Dt_Contagions* shows a behavior similar to the behavior observed in the previous model; the reasons for the number of contagions to have a negative impact on manufacturing imports are due to border closures and government measures in general aimed at containing the accelerated spread of the virus. Hence, as contagions increase, per COVID-19, the logarithm of imports will decrease by $100 \times (-91.60)$:9160 dollars FOB.

The interaction variables *Dt_Deceased* and *Dt_Recovered* have the same behavior as in the previous model, which effectively corroborates that the relaxation of confinement measures and the progressive advance of the pandemic accompany the increase in the number of deaths due to COVID-19. However, although the behavior of the variable *Dt_Recovered*, which considers the relationship with the number of recoveries, also shows a positive relationship between the number of recoveries and imports, as in the previous model, it does not have as great an impact as the number of deaths. Analyzing the coefficients shows that increases in the number of deaths per COVID-19 by one person increase the logarithm of the sector's imports by $100 \times (2406)$: 240,600 dollars FOB; while in increases in the number of recoveries, the logarithm of imports increases by $100 \times (20.31)$:2031 dollars FOB. Table 5 summarizes the impact coefficients.

Conclusions

This research article aims to assess the impact of the arrival of the Covid-19 pandemic in Colombia on the manufacturing industry as a whole. A Difference-in-Differences analysis approach was used to achieve the proposed objective, based on the methodological proposal of natural experiments from a coincidental random and exogenous event aligned with results of interest in the manufacturing industry.

The general consensus is that the crisis has generated a significant economic slowdown in the industries; all this is due to confinement policies, isolation, and

the effects of uncertainty affecting investment plans of industrialists. The economic effects of the Covid-19 pandemic have affected the productive dynamics and the labor market as a whole; the study results show that the arrival of COVID-19 in Colombia had a negative impact on the manufacturing industrial production index in Colombia, with a decline of -11.93571 index points due to this exogenous disturbance. Additionally, it was found that an increase in the interaction between the number of formal employees in the manufacturing sector and the number of COVID-19 infections resulted in a decrease in the index.

When controlling for variables such as remittances, treatment and recovered variables, and the logarithm of the number of long-term unemployed, the results showed a significant negative impact on formal employment in the manufacturing sector after the notification of the first case of COVID-19 in the country, with an average loss of approximately 12.6% of formal workers in the industry. The dynamics of contagions had the expected impact on the logarithm of the number of formal employees in the manufacturing sector, showing a negative relationship due to the increase in COVID-19 infections and the intensification of confinement and isolation measures that led to severe economic contraction. Conversely, the number of people recovered from COVID-19 had a positive impact on the number of formal employees in the industry, as with the progression of the pandemic, containment measures were relaxed for certain productive sectors, generating a positive impact on employment in the industry, as evidenced by the variable estimator.

On the other hand, an important conclusion in this study concerns the statistically significant negative impact on the value of manufacturing exports, indicating a considerable contraction in this sector since its arrival. The government measures to contain the virus spread have led to a decrease in manufacturing sector exports, reflecting the direct impact of confinement on the economy. However, the gradual process of economic reactivation has had a positive impact on manufacturing sector exports. In summary, the recovery of individuals affected by COVID-19 has contributed to resuming productive activities, thereby increasing the flow of foreign trade and exports in the Colombian manufacturing sector.

The analysis reveals that the treatment variable has a negative impact on manufacturing imports, while the economic tracking index (ISE) shows a positive relationship with imports. The contagion interaction indicates that the increase in COVID-19 cases negatively affects imports, possibly due to containment measures. The interaction variables show that the increase in deaths from COVID-19 and recoveries, respectively, have effects on imports, with deaths being the most influential factor.

The models presented clearly demonstrate the negative impact on the economy during the pandemic. It is crucial to consider that a large portion of this sector in Colombia is made up of small and medium-sized enterprises, whose cost structure cannot absorb the additional costs resulting from the decline in sales and the implementation of measures to contain the virus spread. This scenario could lead to the closure of multiple SMEs and cause a significant increase in long-term unemployment with very high social costs.

Furthermore, this study makes a valuable contribution to scientific literature by providing an empirical assessment of the impact of COVID-19, treated as an

exogenous and random experiment. Its innovative methodological approach, using Differences in Differences, offers detailed insights into how the pandemic has affected this critical sector of the economy, thus filling a gap in existing research on the impact of COVID-19 as a natural experiment. The findings highlight the severe economic slowdown experienced by the manufacturing industries due to government policies such as confinement and the uncertainty that has hindered investment plans.

Finally, it is imperative to emphasize the necessity for increased governmental focus on the manufacturing sector, particularly on small and medium-sized enterprises, which are disproportionately affected by the economic repercussions of the pandemic. In conclusion, this study offers invaluable insights for policymakers and stakeholders aiming to alleviate or forecast the detrimental effects of the pandemic, emphasizing the urgency of taking prompt action to tackle these challenges in the days ahead.

Annexes

Table 6 Appendix Shapiro–Wilk W Test for Normal Distribution in Residuals

Model	Variable	Obs	W	V	z	Prob > z
Impact in IPI	U1	20	0.94968	1.191	0.352	0.36229
Impact in Occupied	U2	20	0.97629	0.561	-1.165	0.87779
Impact in Exports	U3	20	0.98131	0.442	-1.644	0.94990
Impact in Imports	U4	20	0.98216	0.422	-1.737	0.9588

Data availability The data that support the findings of this study are available from the corresponding author upon reasonable request.

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